

Disentangling Scale from Targeted Pretraining Depth: Testing the Less-Is-More Hypothesis for Bengali Mathematical Reasoning

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I. INTRODUCTION

Large language models are truly effective at performing mathematical reasoning, but nearly all of this progress has been made for the English language. Despite having more than 230 million speakers, Bengali is considered a low-resource language with respect to the scope of LLM tasks: There are not enough quality Bengali text data in most of the corpora used by pretraining, and there are almost no reasoning-specific data. English-centric models can now solve competition-level math problems, but it is not obvious what it means to have a model which is good for reasoning in Bengali.

Two recent attempts have been made to bridge this gap in opposite directions. TigerLLM (Raihan & Zampieri, 2025) trained a 1B-parameter model on a small (relative to the scale of the generalist models) curated subset of Bengali textbook data, and demonstrated that this outperforms much larger generalist models — suggesting that targeted, high-quality data-pretraining can replace scale. In contrast, TituLLM (Nahin et al., 2025) was pretrained on approximately 37B tokens of general-domain Bengali language data and was found to improve only on select types of tasks. When read side by side, these appear to be contradictory. But they are not actually a clean test of the same question, because in both cases, model size and pretraining depth move together since nobody has held one constant while varying the other. So, it is not really known whether targeted pretraining depth or raw parameter

scale is doing the work, because no prior study has separated the two.

This is the first gap that is addressed in this paper. Even if we can identify what motivates reasoning gains, there are two more questions that arise out of that. First, if a small amount of Bengali math training data does wake up better reasoning, does that improvement stay confined to math, or does it carry over to reasoning more broadly? Second, after a model has developed an improved ability to reason, there is another, and very practical, problem: when has it “reasoned enough?” In addition, any additional token used to “think” results in compute cost, while longer chains of reasoning do not necessarily result in better accuracy, as demonstrated in recent work, sometimes worse. If there were simple, no-training methods of estimating how well a model performs, say, on its own data, then perhaps we could end early without compromising accuracy, but that has not been tested with Bengali specifically.

This paper is organized around three research questions that examine the factors underlying Bengali mathematical reasoning, its generalizability, and the potential for adaptive computation:

RQ1: When a model is provided with a small set of Bengali mathematical examples designed to elicit reasoning, what primarily determines its performance: model scale or the amount of Bengali exposure during pretraining? We investigate this by comparing models that differ in only one of these two factors at a time.

RQ2: When Bengali mathematical reasoning improves, does the improvement generalize beyond mathematics? We evaluate the models on B-REASO, a Bengali exam-style benchmark covering 50 subjects, to determine whether the observed gains transfer broadly or remain concentrated in science-adjacent subjects.

RQ3: Can a model independently determine when it has reasoned sufficiently to stop, without being explicitly trained for this purpose? We compare two low-cost confidence signals and evaluate whether either can reduce computational cost without compromising accuracy.

To answer RQ1, we compare TigerLLM-1B, TituLLM-1B, and TituLLM-3B under an identical small-sample elicitation protocol. Because TigerLLM-1B and TituLLM-1B match in size but differ in pretraining depth, and TituLLM-1B and TituLLM-3B match in pretraining approach but differ in size, this design lets us attribute any performance difference to one factor at a time, something no prior study in this space has been able to do. For RQ2, we evaluate all three post-elicitation models on B-REASO to check whether reasoning gains hold up outside the math domain. For RQ3, we compare a verbalized-confidence approach against an internal-signal approach as candidate stopping criteria, testing both without any additional training.

Together, these three questions target a specific gap in the literature: prior work has looked at scale, at pretraining depth, at elicitation efficiency, and at confidence-based early stopping but always separately, and mostly in English or in settings where scale and depth are not disentangled. This paper brings those threads together for Bengali mathematical reasoning specifically, under a single, controlled comparison.

II. LITERATURE REVIEW

A. The Low-Resource Reasoning Baseline

Reasoning in low-resource languages is a known weak point for LLMs, and Bengali is no exception. Al Nazi et al. (2025) tested six LLMs on four Bengali tasks (news classification, sentiment, summarization, QA) under zero-shot, few-shot, and CoT prompting. The best result, GPT-4o with three-shot CoT, hit 99% F1 on sentiment but CoT was not consistently helpful. In several cases it hurt accuracy, mostly through hallucination, and default hyperparameters turned out to be poorly suited to Bengali across the board. So the failure mode here is not simply “the model isn’t big enough”, it is that reasoning behavior itself is unstable in this setting.

Kadavath et al. (2022) is worth bringing in even though it is English-only, because it gives us the general prior we are testing against: across models from 800M to 52B parameters, self-evaluation and answerability prediction both got more reliable with scale. If that is the general pattern, it raises an obvious question for Bengali specifically, why does the literature keep finding that small, targeted models beat larger generic ones? That is the debate the next section is built around.

B. The Backbone Debate: Scale vs. Targeted Depth

This is the central disagreement in the literature, and it splits into three positions rather than a clean two-sided debate.

The **targeted pretraining** side is TigerLLM (Raihan & Zampieri, 2025). It was pretrained on only ~ 10 M tokens of Grade 6–12 Bengali textbook content plus 100k instruction pairs distilled from GPT-4o and Claude, yet its 1B-parameter

variant outperformed 11B generalist models, reaching 68% accuracy on BanglaRQA. The claim is not just “small models can be competitive.” It is that targeted depth can substitute for scale outright.

The **scale universalist** side pushes back. Yong et al. (2025) found that test-time scaling only becomes reliable at ≥ 3 B parameters, with a 14B model sitting at a kind of sweet spot, though even there, generalization to cultural, out-of-domain knowledge stayed weak, and Bengali’s tokenizer made reasoning about 3.5x more expensive than English. MathMist (Sobhani et al., 2026), a 30k-pair benchmark across 13 languages, backs this up: scaling Qwen from 0.6B to 14B shrank the Bengali-English accuracy gap from 7% down to 1%, and smaller models showed what the authors call linguistic brittleness during CoT.

A third position complicates both of these. TituLLM (Nahin et al., 2025) pretrained on ~ 37 B tokens of general-domain Bengali with an extended tokenizer, and found that targeted Bengali pretraining helped with commonsense-style tasks but did not help with broad-knowledge tasks like MMLU, an effect they partly attribute to only training for one epoch and skipping instruction tuning. So the scale-vs-depth question is not binary. It looks like it depends on what kind of reasoning you are asking for.

The problem across all three papers is that none of them actually isolates scale from pretraining depth, TigerLLM’s depth claim only exists at 1B/9B, and TituLLM’s scale claim only exists within its own general-corpus setup, so the two variables are always tangled together. That is the gap this paper is built to close: fixing scale at 1B while varying pretraining depth (TigerLLM-1B vs. TituLLM-1B), and fixing the pretraining approach while varying scale (TituLLM-1B vs. TituLLM-3B), so the two effects can finally be told apart.

Even if we accept that scale and depth interact to determine a model’s underlying capacity, there is a separate question: how much elicitation data does it actually take to surface that capacity in the first place?

C. The Less-Is-More Elicitation Paradigm

A second, mostly separate strand of the literature argues that eliciting reasoning in an already-capable model takes surprisingly little data which is really the whole justification for using a small set of Bengali math examples as the intervention here, rather than treating it as an underpowered shortcut.

LIMO (Ye et al., 2025) makes this point cleanly: 800 curated reasoning traces elicited more sophisticated reasoning than training on the 100k-sample NuminaMath dataset, with gains only saturating around 1.6k samples or 32B parameters. s1 (Muennighoff et al., 2025) gets to a similar place through a different route — just 1k reasoning traces plus “budget forcing” (inserting “Wait” tokens to stretch out inference-time reasoning) let s1-32B beat o1-preview on AIME24 by 27%, and sequential scaling outperformed parallel sampling.

LESS (Xia et al., 2024) gives a plausible reason *why* this works. Selecting just 5% of a 270k-point pool, using projected low-rank gradients to find the most functionally relevant

examples, beat training on the full dataset. The takeaway is that functional relevance beats raw volume which lines up with the logic behind TigerLLM’s small, curriculum-aligned corpus in Section 2.2.

This section is not a competing claim to Section 2.2 so much as its scaffolding: it is what makes small-sample elicitation a reasonable thing to test at all. The catch is that almost all of this evidence comes from English or other high-resource settings. Whether elicitation stays this efficient once the reasoning itself has to happen in a low-resource language is a different question and one the next section complicates.

D. The Elicitation Tax: Native Reasoning vs. English-Pivot

There is a real cost attached to reasoning in Bengali specifically, and it cuts against both the training setup in Section 2.3 and the generalization question that follows.

Qi et al. (2025), using the 370-question, 11-language XReasoning benchmark, found that forcing a model to reason natively in Bengali instead of pivoting through English dropped accuracy on hard tasks like AIME from 26% to 17%. Their proposed fix was tested on at most 817 samples, so it’s a partial answer at best. Yong et al. (2025) adds to this from a different angle — Bengali’s tokenizer makes reasoning about 3.5x more expensive, and even where in-domain scaling works, generalization to out-of-domain cultural knowledge stays weak.

There’s a related but different problem with distillation. BenNumEval (Ahmed et al., 2025), across 3.2k samples in six numerical reasoning categories, found that DeepSeek-R1-Distill-Llama-70B — despite being built from top-tier English reasoning distillation — collapsed to 4.7% accuracy in native Bengali, well below Gemini-2.0-flash’s 79.6% and even below several non-distilled baselines. That’s not just a failure to transfer. It suggests distilling English reasoning traces can actively damage a model’s existing multilingual knowledge.

SOMADHAN (Paul et al., 2025) offers an important qualifier here. Using 8.8k expert-translated Bengali word problems to elicit native CoT directly, it worked — Llama-3.3-70B hit 88% — but smaller models mostly failed to adapt to native-language reasoning at all. So the elicitation tax doesn’t seem to apply evenly; it interacts with model size, which matters a lot for a 1B-vs-3B comparison.

Given this tax, the next question is whether whatever a small Bengali training set produces actually carries beyond math, or stays locked to the domain it was trained on.

E. The Generalization Gap

This section is short on purpose. None of the sources reviewed here test whether math-elicited reasoning generalizes to a broad, multi-subject exam-style benchmark — nothing like B-REASO’s 50-subject spread shows up anywhere in this literature.

The closest things we have are indirect. TituLLM’s System-1/System-2 split (Section 2.2) hints that pretraining-driven gains might be capability-specific rather than general. The explainability- and distillation-tax findings (Section 2.4) show

that domain-specific training can sometimes actively hurt broader multilingual ability rather than just fail to improve it. Neither one is a direct test of cross-domain transfer, but both point the same direction — that gains staying narrow to math is a real possibility, not a strawman.

So RQ2 is really addressing a gap in the evidence, not settling an existing disagreement. There’s nothing here to confirm or contradict — just enough adjacent signal to say the question is genuinely open.

Separate from what gets elicited and whether it generalizes, there’s a third question running through this literature: once a model is reasoning, how does it know when to stop?

F. The Scaling Paradox: Thinking Longer vs. Overthinking

There’s a real contradiction in the test-time compute literature over whether longer reasoning chains help or hurt.

s1 (Muennighoff et al., 2025) and LIMO (Ye et al., 2025) (both from Section 2.3) fall on the “longer helps” side — budget forcing lets a model catch and fix its own early mistakes, and this shows up as accuracy gains of up to 7% on competitive math benchmarks.

Other papers find close to the opposite. Zeng et al. (2025), across MATH, AIME, Omni-MATH, and GPQA, found that correct answers were consistently *shorter* than incorrect ones, and that reasoning rarely converts a wrong answer into a right one — the wrong-to-correct revision rate stayed under 10%. (Worth noting: this leaned heavily on distilled models rather than the full R1-671B, and only looked at static checkpoints.) TOPS (Yang et al., 2025), built on 1.3k seed problems and 26k self-improvement samples, found that pushing CoT length too far actually hurt accuracy on easy problems specifically, and that models tend to converge on correct answers at some “optimal” — not maximal — amount of reasoning. That was tested only in the SFT setting and mostly in math.

Two more papers add nuance without resolving the disagreement. Snell et al. (2024) showed that allocating compute based on problem difficulty (search vs. revision) beat a model 14x larger on easy-to-medium problems, with revision working best on the easy end — but this depended on “oracle” difficulty labels that aren’t actually available at inference time, and the results were limited to PaLM-2 on MATH. Atom of Thoughts (Teng et al., 2025) proposed breaking reasoning into Markovian steps, each independent of the full history, cutting overhead — though this hasn’t been tested on open-ended or non-verifiable tasks.

A field-level survey backs up this split rather than resolving it. Chen et al. (2026), reviewing the long-CoT literature broadly, confirm that the overthinking phenomenon is real and reproducible across studies, while also arguing that sequential test-time scaling is fundamentally bounded by a model’s internal reasoning limits — meaning that past some point, letting a model think longer stops helping regardless of how the stopping decision is made.

Put together, the field seems to agree on one thing: length alone isn’t a reliable stand-in for reasoning quality. Where it

splits is on what should replace it — which is exactly what the confidence literature tries to answer.

G. The Confidence Conflict: Verbalized vs. Probed Uncertainty

Two different schools propose two different “cheap” ways to tell a model when to stop, and they disagree on where that signal should come from.

One camp says the signal can just come from what the model says. MetaFaith (Liu et al., 2025), across ten datasets and six domains, found that metacognitive prompting improved calibration — how well a model’s stated confidence matches its actual accuracy — by 61%. Interestingly, optimizing for factual accuracy of confidence statements can actually hurt this faithfulness, and the whole study was English-only. Kadavath et al. (2022), again, adds that while self-evaluation improves with scale, zero-shot verbalized confidence on its own isn’t well-calibrated without extra help.

The other camp says the model’s internal state is a better signal than anything it says out loud. Thought Calibration (Wu et al., 2025), on a 1k-question math/science split, trained linear probes on the model’s last-layer hidden states to decide when to stop reasoning — cutting thinking tokens by 60% with no drop in accuracy. Probes based on consistency generalized best out-of-distribution, though they can only stop reasoning, not correct it, and depend on how similar the calibration set is to the actual task. CaTS (Huang et al., 2025), using just 2k questions, distilled self-consistency signals directly into the model’s output logits and saved 94.2% of the sampling budget at the same accuracy target — beating vanilla self-consistency, with the same calibration-similarity caveat.

Both camps share a limitation that matters here: each either needs extra training (probes, distillation) or has only been validated in English. Neither has been tried as a genuinely training-free, Bengali-native stopping signal — which is exactly the comparison RQ3 sets out to make.

H. Gap Statement

Across this literature, three gaps stand out, one for each RQ.

RQ1. No Bengali-language study separates parameter scale from pretraining depth as independent factors — the studies in Section 2.2 always vary size and corpus design together, so it’s impossible to tell from existing work which one is actually doing the work. It’s also untested whether small-sample elicitation, shown to work well in high-resource settings (Section 2.3), still holds up once the native-language reasoning tax (Section 2.4) is factored in.

RQ2. No source tests whether reasoning gains from math training carry over to a broad, multi-subject benchmark rather than staying confined to the training domain (Section 2.5).

RQ3. No source tests cheap, training-free confidence signals — verbalized against internally probed — specifically for deciding when to stop reasoning in Bengali, even though both approaches have been separately validated in English (Sections 2.6–2.7).

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ACKNOWLEDGMENT

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